

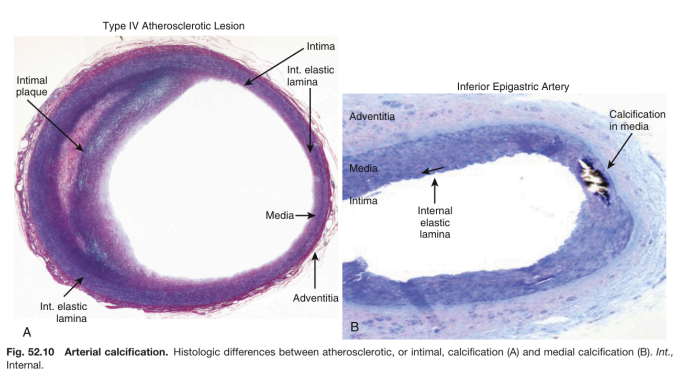


Fig. 52.10 Arterial calcification. Histologic differences between atherosclerotic, intimal, calcification (A) and medial calcification (B). Int., internal.

Fig_52_10

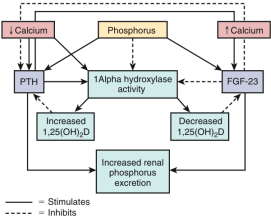


Fig. 52.13 Regulation of serum phosphorus levels. As phosphorus levels increase (or there is a long-term phosphorus load), levels of both parathyroid hormone (PTH) and fibroblast growth factor 23 (FGF-23) are increased. Both of these elevations in turn increase urinary phosphate (Pi) excretion. The two hormones differ in respect to their effects on the vitamin D axis. PTH stimulates 1α-hydroxylase activity, thereby increasing the production of calcitriol, which in turn negatively feeds back on the parathyroid gland to decrease PTH secretion. By contrast, FGF-23 inhibits 1α-hydroxylase activity, thereby decreasing the production of calcitriol, thus feeding back to stimulate further secretion of FGF-23. FGF-23 and PTH also regulate each other. Finally, low calcium levels stimulate PTH, whereas high calcium levels stimulate FGF-23. Lastly, there is some evidence that FGF-23 also inhibits PTH secretion.

Fig_52_13

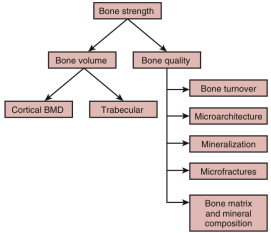


Fig. 52.16 Determinants of bone strength. Bone strength comprises both bone density and bone quality. "Bone quality" refers to bone turnover, microarchitecture, microfractures, mineralization, and composition of the mineral matrix. "Trabecular microarchitecture" involves trabecular thickness, the ratio of plates and rods, and their connectivity and spacing. "Cortical microarchitecture" consists of cortical thickness, porosity, and bone size. Composition of the mineral matrix includes changes in the cross-linking of type I collagen and alterations in the size and structure of bone mineral. Bones accumulate microfractures over time even with normal physical activity. The ability to repair them affects bone quality. *BMD*, Bone mineral density. [From Moorthi R, Moe S. Recent advances in the noninvasive diagnosis of renal osteodystrophy. *Kidney Int*. 2013;84:899–894. With permission.]

Fig_52_16

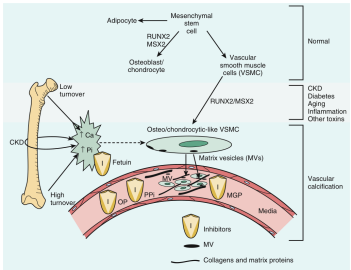


Fig. 52.11 Overview of the pathophysiology of vascular calcification. Normally, mesenchymal stem cells differentiate to adipocytes, osteoblasts, chondrocytes, and vascular smooth muscle cells (VSMCs). In the setting of chronic kidney disease (CKD), diabetes, aging, inflammation, and the presence of multiple other factors, these VSMCs can dedifferentiate or transform to chondrocyte/osteoblast-like cells by upregulation of transcription factors such as runt-related transcription factor 2 (*Runx2*) and homeobox protein *Msx2*. These transcription factors are critical for normal bone development, and thus their upregulation in VSMCs is indicative of a phenotypic switch. These osteoblast-like VSMCs then become calcified in a process similar to bone formation. The cells lay down collagen and noncollagenous proteins in the intima or media and incorporate calcium (Ca) and phosphorus (Pi) into matrix vesicles to initiate mineralization and further grow the mineral into hydroxyapatite. The overall positive calcium and phosphorus balance of most patients undergoing dialysis feeds both the cellular transformation and generation of matrix vesicles (MVs). In addition, the extremes of low turnover in CKD (low and high turnover or adynamic and hyperparathyroid bone, respectively) increase the available calcium and phosphorus by altering the bone content of these minerals. Ultimately, whether an artery calcifies or not depends on the strength of the array of inhibitors (a) standing by in the circulation (fetuin-A) and in the artery (e.g., pyrophosphate [PPi], matrix glycoprotein [MGP], and osteopontin [OPN]). [From Moe SK, Chen NX. Mechanisms of vascular calcification in chronic kidney disease. *J Am Soc Nephrol*. 2008;19:213–216. With permission.]

Fig_52_11

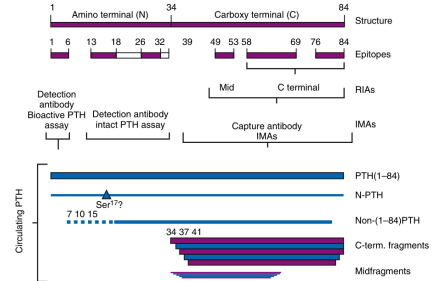


Fig. 52.14 Parathyroid hormone (PTH) assays. Schematic presentation of PTH(1–84) and the relationship between PTH assays. PTH assay epitopes, and PTH molecular forms detected in the circulation. The upper panel depicts the structure of human PTH and the epitopes detected by various PTH assays. First-generation PTH assays detect full-length PTH(1–84) in addition to PTH fragments. These assays include radioimmunoassays (RIA) that use antisera specific to the amino-terminal (N-RIA), middle (M-RIA), or carboxy-terminal (C-RIA) regions of PTH. Second-generation "intact PTH" assays detect full-length PTH(1–84) and non-(1–84)-PTH fragments. Third-generation PTH assays (bipoint PTH) detect only full-length PTH(1–84). The bottom panel depicts PTH molecular forms present in the circulation. IMA, Immunomimetic assay; N-PTH, amino-terminal PTH; term., terminal. [From Heinrich LM, Rogul AD, O'Anur T, et al. Persistent hypercalcaemia after parathyroidectomy in an adolescent and effect of treatment with calcitonin. *HLJ Clin Chem*. 2006;52:2286–2293.]

Fig_52_14

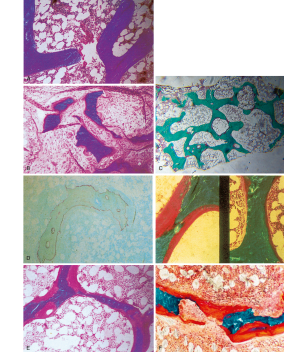


Fig. 52.17 Bone histology. (A) Normal bone. (B) Bone tissue with increased osteoclast activity and decreased osteoblast activity. (C) Bone tissue with increased osteoclast activity and decreased osteoblast activity. (D) Bone tissue with increased osteoclast activity and decreased osteoblast activity. [From Moorthi R, Moe S. Recent advances in the noninvasive diagnosis of renal osteodystrophy. *Kidney Int*. 2013;84:899–894. With permission.]

Fig_52_17

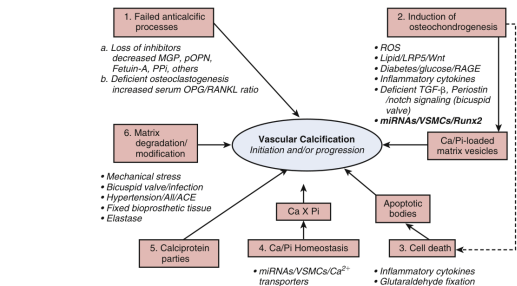


Fig. 52.12 Factors that regulate pathways involved in the pathogenesis of vascular calcification. Multiple factors regulate each step of extracellular calcification. ACE, Angiotensin-converting enzyme; Ang, angiotensin II; Ca, calcium; LRP5, low-density lipoprotein (LDL) receptor-related protein 5; MGP, matrix gamma-carboxyglutamate (Gla) protein; miRNA, microRNA; OPN, osteopontin; P, phosphate; pOPN, osteopontin; PPI, pyrophosphate; RAGE, advanced glycosylation and product receptor; RANKL, receptor activator of nuclear factor κB ligand; ROS, reactive oxygen species; Runx2, runt-related transcription factor 2; TGF-β, transforming growth factor β; VSMCs, vascular smooth muscle cells; Wnt, wingless-type MMTV integration site family member. [From Wu M, Remmer C, Giachelli CM. Vascular calcification: an update on mechanisms and challenges in treatment. *Calcif Tissue Int*. 2013;93:365–373. With permission.]

Fig_52_12

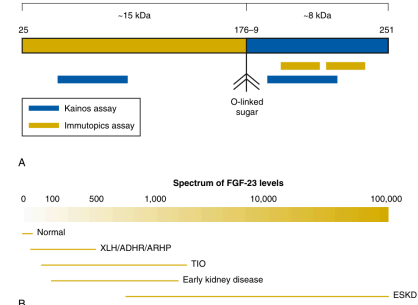


Fig. 52.15 Fibroblast growth factor 23 (FGF-23) assays. (A) FGF-23 O-glycosylation site and epitopes recognized by antibodies used in current assays. (B) Spectrum of serum FGF-23 levels in early chronic kidney disease and end-stage kidney disease (ESKD) compared with the normal reference range and levels associated with different disorders affecting FGF-23. *ADHR*, Autosomal dominant hypophosphatemic rickets; *ARHP*, autosomal recessive hypophosphatemia; *TIO*, tumor-induced osteomalacia; *XLH*, X-linked hypophosphatemia. [From Block GA, et al. Kettler M, et al. Phosphate homeostasis in CKD: report of a scientific symposium sponsored by the National Kidney Foundation. *Am J Kidney Dis*. 2013;62:457–473. With permission.]

Fig_52_15

Table 52.1 Kidney Disease Improving Global Outcomes Classification of Chronic Kidney Disease–Mineral Bone Disorder and Renal Osteodystrophy	
Definition of chronic kidney disease–mineral bone disorder (CKD-MBD)	A systemic disorder of mineral and bone metabolism due to CKD manifested by one or a combination of the following: Abnormalities of calcium, phosphorus, parathyroid hormone, or vitamin D metabolism Abnormalities in bone turnover, mineralization, volume, linear growth, or strength Vascular or other soft tissue calcification
Definition of renal osteodystrophy	Renal osteodystrophy is an alteration of bone morphology in patients with CKD. It is one measure of the skeletal component of the systemic disorder of CKD-MBD that is quantifiable by histomorphometry of bone biopsy.
From Moe S, Drüeke T, Cunningham J, et al. Definition, evaluation, and classification of renal osteodystrophy: a position statement from Kidney Disease: Improving Global Outcomes (KDIGO). <i>Kidney Int</i> . 2006;69:1945–1953.	

Table_52_1

Table 52.2 Comparison of Phenotypes of Klotho Deficiency and Chronic Kidney Disease		
	Klotho Deficiency	Chronic Kidney Disease
Blood Chemistry		
Phosphate		I or ^a
Calcium	I	— or II
Creatinine	I	
Calcitriol		
Parathyroid hormone	— or I	
Fibroblast growth factor 23		
Klotho	or disappears	at ESKD ^b
Gross Phenotypes		
Body weight		
Growth retardation		in children
Physical activity		I
Fertility		
Life span		
Cardiovascular Disease		
Cardiac hypertrophy		
Cardiac fibrosis		
Vascular calcification		
Atherosclerosis		
Blood pressure	I	
Hemostatic levels	I	
Bone disease		
^a During early chronic kidney disease, blood phosphate level is in the normal range. ^b Blood klotho may be increased in early chronic kidney disease. ESKD, End-stage kidney disease; I, decrease; II, increase; —, unchanged. Modified from Hu MC, Kuro-o M, Moe OW. Parathyroid and extrarenal actions of klotho. <i>Semin Nephrol</i> . 2013;33:118–129.		

Table_52_2

Table 52.3 Biomarkers for Chronic Kidney Disease–Mineral Bone Disorder								
	Affected by Sample Processing	Assay Validity	Renally Excreted	Diurnal Variation	Seasonal Variation	Variation With Meals	Variation With Dialysis Time	
Parathyroid hormone	Yes	No; some assays pick up fragments	No	Yes	No	No	No	Yes
25(OH)D (calcitriol)	No	Good (uncertain importance of differentiating D ₂ from D ₃)	No	No	Yes	No	No	No
1,25(OH) ₂ D (calcitriol)	No	Good	No	Yes	No	No	?	?
Fibroblast growth factor 23	No	Intact versus C terminal	No	?	?	Yes	No	No
Soluble α-klotho	?	Uncertain	Yes	?	?	?	?	?
Sclerostin	?	Uncertain, likely valid	No	?	?	?	?	?
Bone-specific alkaline phosphatase	No	Good	No	No	No	?	No	No
Assay validity indicates that the measurement is of the biologically active hormone or marker, not fragments. ?, Indicates insufficient data.								

Table_52_3

Table 52.4 Causes and Proposed Mechanisms of Decreased Bone Formation in Patients With Chronic Kidney Disease	
	Mechanism of Decreased Osteoblast Activity
Low serum calcitriol	I Osteoblast differentiation
Metabolic acidosis	I Osteoblast life span
	I Calcitriol production
High serum phosphate	I Collagen synthesis
Calcium loading	I Calcitriol production
Hyperparathyroidism	I Calcitriol production and osteoblast degradation
	I Osteoblast for calcium-sensing receptor
High serum interleukin-1 and 6, tumor necrosis factor	I Osteoblast life span
Low serum insulin-like growth factor-1 (IGF-1) activity	I IGF-1 and IGF-binding protein 1 (IGFBP1) levels
	I Inhibitory IGF-2, 4, 6 levels
Sclerostin	I Osteoblast life span
	I Wnt1/catenin signaling
Dishevelled-1 (pks-1)	I Osteoblast activity
Mineralization, osteonema	I Wnt1/catenin signaling
	I IGF-1, 25-hydroxyvitamin D levels
Diabetes	I IGF-1, 25-hydroxyvitamin D and calcitriol levels
	I Osteoblast degradation and products (AGEs)
	I Osteoblast life span
	I AGEs
Age-related	I Osteoblast life span
Hypogonadism	
Women (I estrogen and I sex hormone-binding globulin [SHBG])	I Osteoblast life span
Men (I testosterone and I SHBG)	I Osteoblast life span
Uremic toxins (urea acid)	I Calcitriol production
	I Vitamin D receptor activity
Aluminum toxicity	I Osteoblast proliferation
	I Osteoblast activity
I, Decreased; II, increased.	

Table_52_4

Table 52.5 Turnover, Mineralization, Volume Classification System for Renal Osteodystrophy

Turnover	Mineralization	Volume
Low	Normal	Low
Normal		Normal
High	Abnormal	High

From Moe S, Drüeke T, Cunningham J, et al. Definition, evaluation, and classification of renal osteodystrophy: a position statement from Kidney Disease: Improving Global Outcomes (KDIGO). *Kidney Int*. 2006;69:1945–1953.

Table_52_5